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Using Post Stack Deconvolution to Enhance the Utilization of Spectral Decomposition in Stratigraphic Interpretation

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Abstract

The effectiveness of spectral decomposition in seismic interpretation largely depends on the bandwidth and amplitude of frequencies present in the seismic volume. In deeper reservoir intervals, high-frequency signals tend to be attenuated, resulting in spectral volumes with limited resolution and weak stratigraphic detail. This study introduces a streamlined post-stack deconvolution workflow designed to enhance the spectral content of seismic data prior to decomposition. By extracting and averaging wavelets from well log data and applying deconvolution trace-by-trace, the embedded wavelet is replaced with a more realistic representation of the subsurface response. This improves the frequency bandwidth and enables a more detailed spectral decomposition. Comparative analysis between the original and deconvolved volumes—visualized through RGB spectral blends—demonstrates that the deconvolved data exhibits a broader frequency range (up to 75 Hz vs. 60 Hz), stronger amplitudes in the mid-to-high frequency bands, and greater detail, particularly in deeper sections. New stratigraphic features become visible, and previously subtle features are rendered with improved clarity. This approach offers a simple yet effective enhancement to seismic interpretation workflows, requiring minimal computational effort and no reprocessing of the original seismic data.

Introduction

Seismic data serves as the cornerstone of subsurface interpretation, enabling geoscientists to delineate structural features, stratigraphic architectures, and reservoir properties. The resolution of this data, however, is inherently limited by the characteristics of the seismic wavelet and the band-limited nature of the seismic acquisition and processing. These limitations become especially pronounced in deeper zones, where attenuation reduces the amplitude and clarity of higher frequency reflections.

To address these challenges, interpreters increasingly rely on data conditioning techniques to boost the effectiveness of frequency-based interpretation tools. Among these, post-stack deconvolution stands out as a low-cost, high-impact enhancement, particularly when integrated with workflows like spectral decomposition. While conventional processing routines address deconvolution early in the workflow, applying it post-stack using well-derived wavelets allows for greater control, validation, and interpretive

feedback. This approach bridges the gap between data conditioning and geological insight, helping geoscientists enhance features that were previously masked or attenuated by wavelet effects.

Additionally, as seismic data acquisition and processing becomes more complex and costlier, maximizing the interpretive value of already processed volumes is critical. Simple, effective enhancements such as the application of deconvolution can be a cost-effective alternative to full reprocessing. This not only saves time and resources but also aligns with the growing industry trend of lean, data-driven exploration techniques.

One critical component in overcoming these challenges is spectral decomposition, a method that analyzes seismic data in the frequency domain to reveal subtle geological features such as thin beds, stratigraphic traps, and changes in lithology. However, spectral decomposition itself is dependent on the fidelity of the input seismic signal. If the seismic data are dominated by low-frequency content or suffer from wavelet effects, spectral attributes may appear bland, obscuring valuable subsurface features.

This study aims to address that limitation by introducing a quick, practical method to enhance seismic resolution using post-stack deconvolution prior to spectral decomposition. By removing the wavelet from the seismic trace using well-derived data, the reflectivity series becomes more informative, allowing higher frequencies to be more effectively analyzed. This workflow not only improves the interpretability of deep geological layers but also provides a higher-resolution basis for frequency-based stratigraphic interpretation. The method proposed in this study is particularly suited for large-scale regional studies or legacy datasets where reprocessing is not feasible but interpretive resolution is still required, however, in this study it was mainly tested on a small area to identify lithologies and stratigraphic traps.

The proposed workflow comprises five primary stages: (1) well tie and wavelet extraction, (2) wavelet averaging and design of the deconvolution operator, (3) application of post-stack deconvolution to the seismic volume, (4) frequency analysis and selection of frequency bands, and (5) spectral decomposition and RGB blending for comparison.

Background and Motivation

Literature Review

Several studies have addressed the challenges of improving resolution in seismic data through post-stack processing techniques. [Yilmaz \(2001\)](#) emphasized the importance of deconvolution in enhancing temporal resolution and recovering the reflectivity series. [Partyka et al. \(1999\)](#) introduced spectral decomposition as a method for mapping stratigraphic features based on frequency content, showing that its effectiveness relies heavily on the input data's frequency spectrum.

Further research by [Castagna et al. \(2003\)](#) and [Marfurt and Kirlin \(2001\)](#) demonstrated that combining spectral decomposition with high-resolution seismic data significantly improves interpretation of depositional environments. However, they also noted that without pre-conditioning steps such as deconvolution, frequency-domain analysis can be limited by the band-limited nature of seismic wavelets. This study builds upon these findings by integrating a quick, wavelet-informed deconvolution prior to spectral analysis and spectral decomposition.

Spectral decomposition techniques have been widely used in the oil and gas industry for decades to identify thin beds, channel systems, and other stratigraphic features. However, the effectiveness of these techniques is inherently limited by the frequency content of the input seismic data. Most seismic data is band-limited, and due to attenuation mechanisms like absorption and scattering, high frequencies decay faster with depth. As a result, deep reflectors often appear smeared and lack the clarity required for high-resolution interpretation.

Deconvolution, particularly post-stack deconvolution, is a method aimed at compressing the seismic wavelet and thereby increasing the temporal resolution of seismic data. While typically used during the processing stage, applying a well-designed deconvolution in the interpretation phase using a wavelet derived

from well data can effectively enhance the seismic bandwidth. This approach is not new but has not been extensively combined with spectral decomposition workflows, especially in a post-stack context.

Our motivation is to fill this gap and provide a practical enhancement to the spectral decomposition process that increases the interpretability of deeper stratigraphic features, often neglected due to their poor resolution.

Regional Geological Setting of the Study Area

The study area includes multiple hydrocarbon fields that feature distinct types of reservoirs. In the shallow subsurface, carbonate reservoir dominates, while deeper sections are primarily composed of clastic reservoirs. This geological setting reflects a diverse and complex stratigraphic framework. The region has experienced a long and intricate geological history involving various tectonic events, sedimentation cycles, and diagenetic processes. These geological influences have played a crucial role in shaping the structural and stratigraphic configurations of the area, thereby affecting the distribution, quality, and productivity of hydrocarbon reservoirs. As a result, understanding this complexity is essential for effective exploration and development strategies.

Tectonic Framework

The structural configuration of the study area is predominantly shaped by the Hercynian Orogeny, a significant tectonic event during the Late Carboniferous to Early Permian periods. This orogeny led to the formation of north-south trending basement-involved horst blocks. These horsts are evident as positive gravity anomalies and have played a crucial role in hydrocarbon accumulation.

Stratigraphy and Reservoirs

The study area fields are primarily characterized by high-quality Jurassic-age carbonate reservoirs. Furthermore, clastic reservoirs are present in deep stratigraphic levels, indicating a multi-reservoir petroleum system.

These reservoirs are structurally controlled by elongated, north–south trending, asymmetrical anticlines, which are a result of regional tectonic deformation. Notably, the upper stratigraphic zones generally exhibit superior reservoir properties—such as increased porosity and permeability—compared to the deeper intervals, enhancing their effectiveness in hydrocarbon storage and production.

Implications for Seismic Interpretation

Understanding the regional geological setting is vital for seismic interpretation and reservoir characterization. The complex stratigraphy and structural features influence seismic wave propagation, affecting the quality of seismic data. Incorporating geological insights into seismic processing, such as post-stack deconvolution, can enhance the resolution of seismic images, particularly in deeper stratigraphic intervals.

Methodology

Seismic Acquisition and Processing

Seismic acquisition initially focused on the northern sector of the Study Area. Subsequently, the acquisition efforts were extended to encompass the southern portion of the complex, ensuring full spatial coverage across multiple fields and areas of interest. The entire seismic survey was carried out using conventional 3D acquisition.

Software Tools and Platforms

All wavelet extraction and well-to-seismic ties were conducted using a platform that provided robust capabilities for synthetic seismogram generation, correlation, and wavelet averaging across multiple wells. This ensured a consistent and regionally representative wavelet for subsequent processing. The deterministic deconvolution step was performed using a software tool that leveraged an in-house developed plug-in to deconvolve the wavelet trace by trace. Finally, another application was used for spectral decomposition, offering high-resolution RGB frequency blends and a user-friendly visualization interface. This suite enabled seamless integration of the workflow from wavelet extraction to spectral interpretation, allowing for efficient evaluation and comparison of results.

Seismic Deconvolution Explained

Seismic deconvolution is a signal processing technique aimed at improving the temporal resolution of seismic data. The fundamental concept behind deconvolution is to reverse the effects of the seismic wavelet, which acts as a band-limited filter on the reflectivity series recorded during seismic acquisition.

In seismic data, the recorded signal is the convolution of the earth's reflectivity with the seismic source wavelet. Since the wavelet is typically of limited bandwidth and contains side lobes, it smears the reflectivity, making individual reflection events harder to distinguish. Deconvolution attempts to remove or compress this wavelet effect, ideally retrieving a more spike-like reflectivity series, which allows for better identification of geological boundaries.

There are several types of deconvolution techniques, including predictive deconvolution, spiking deconvolution, and wavelet deconvolution. In the context of this study, we focus on wavelet deconvolution, where a wavelet extracted from well ties is used to invert the seismic traces. This approach allows for the removal of the actual estimated wavelet from the seismic signal, rather than relying on statistical assumptions.

By enhancing the resolution, especially in the vertical direction, deconvolution plays a crucial role in highlighting subtle features that may be obscured by the wavelet's effects. This is particularly beneficial in spectral decomposition, where the presence of a cleaner reflectivity series enables the capture of higher-frequency components that are otherwise attenuated or masked.

Well Tie and Wavelet Extraction / Averaging

The first step in the process involves tying the wells in the area of interest to the seismic volume. Synthetic seismograms are generated using sonic and density logs, and these are correlated with the seismic traces at the well location to extract an accurate wavelet. This extracted wavelet represents the convolutional operator that has blurred the reflection series in the seismic data.

Multiple wells across the area are used to extract wavelets, which are then compared and averaged to create a representative wavelet for the entire survey. This averaging helps reduce local variability and ensures that the deconvolution process does not overfit to a single well's response.

Wavelet extraction windows were selected to span from the shallow section to the deeper section of the seismic data in order to evaluate how deconvolution performance varies across different depths. This approach facilitated a comparative analysis of the deconvolution's effectiveness in different intervals. However, for optimal results it is generally advisable to extract the wavelet specifically within the target zone or area of geological interest, where the interpretation is focused.

Application of Post-Stack Deconvolution

The designed deconvolution operator is then applied to every trace in the seismic volume. This step significantly enhances the temporal resolution and boosts the amplitude of higher frequencies that were previously suppressed. While this does not fully recover lost high frequencies due to attenuation, it improves the relative balance and visibility of existing mid-to-high frequency components in the data.

We observe a noticeable sharpening of events and a better delineation of reflectors, especially in deeper sections. This enhancement forms the basis for the improved performance of spectral decomposition in subsequent steps.

Frequency Analysis and Band Selection

After deconvolution, frequency analysis is conducted on both the original and deconvolved seismic volumes. Spectral analysis tools are used to examine the amplitude spectra of traces across various depth intervals.

Based on this, three frequency bands are chosen for the spectral decomposition—typically low (<20 Hz), mid (20–40 Hz), and high (>40 Hz)—to construct RGB blends. These frequency slices are computed using constant-frequency spectral decomposition techniques.

Spectral Decomposition and Visualization

The selected frequency bands are then used to create RGB blends for both the original and deconvolved volumes. Additionally, the volumes are flattened along a regional horizon marker to better visualize stratigraphic changes and depositional features.

The RGB blends, original and deconvolved, are compared side-by-side in both shallow and deep sections. The differences are quantified using frequency amplitude histograms, interpretive evaluation by expert interpreters, and comparison between the RGB blends and well data.

Results and Observations

The results of the workflow clearly demonstrate the benefits of applying post-stack deconvolution prior to spectral decomposition. Several key observations are summarized below:

The analysis reveals that while the original data has a dominant frequency band up to about 60 Hz, the deconvolved data shows significant energy up to 75 Hz, with better amplitude retention in the 35–75 Hz range. (Figure. 1) These observations inform the selection of frequency bands for the RGB spectral decomposition. This directly translates to better resolution in the spectral decomposition results.

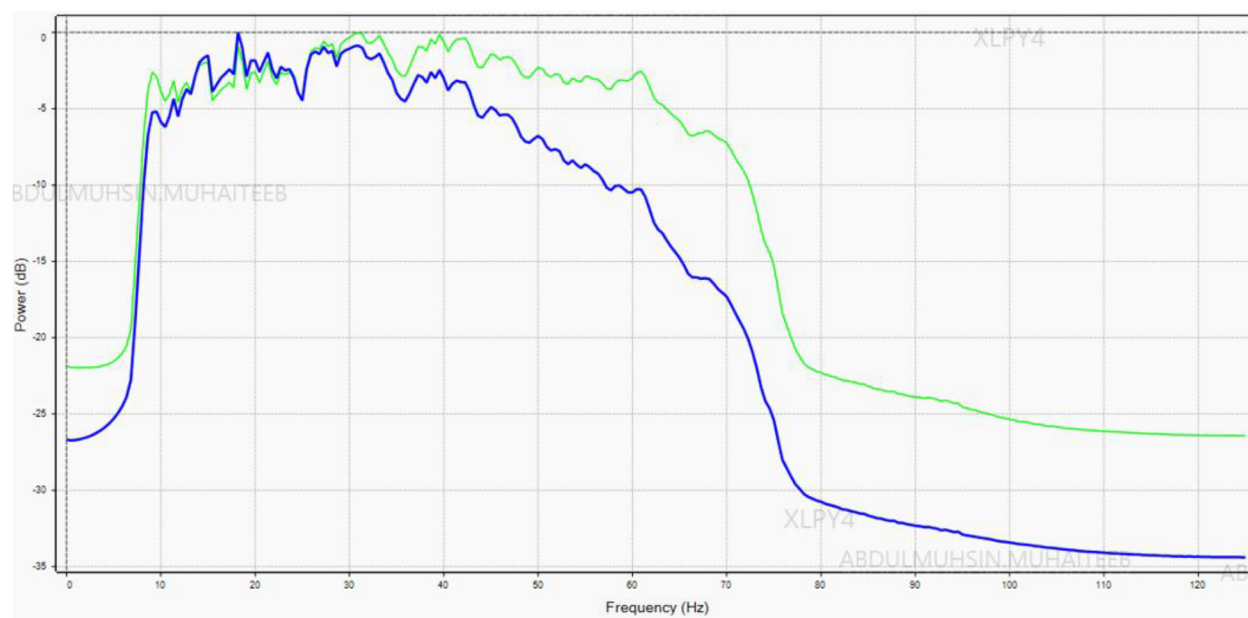


Figure 1—Frequency spectrum of original volume (blue) vs frequency spectrum of deconvolved volume (green)

In the shallower section of the seismic volume, both the original and deconvolved RGB blends show broadly similar features, although the deconvolved version appears slightly sharper. (Figure. 2) However, the benefits become much more apparent in the deeper sections. The original volume's spectral blend is dominated by low-frequency energy (red) and lacks detail in the deeper section. In contrast, the deconvolved volume reveals previously hidden features, including subtle channels, stratigraphic traps, and possible onlap surfaces. These features are not only visible but are more continuous and better defined than seen in the original volume.

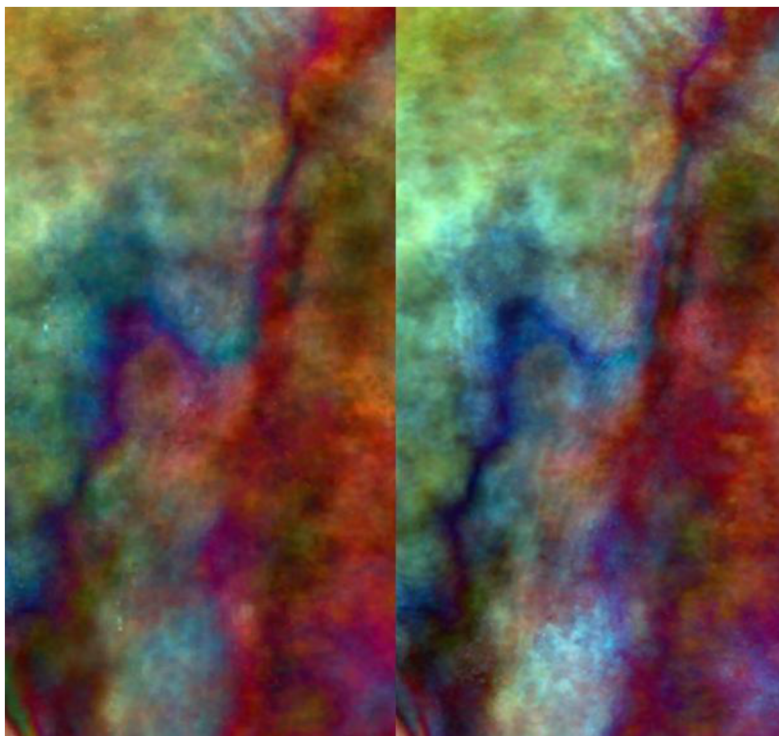


Figure 2—Stratigraphic feature is clearer and more continuous in the deconvolved SD blend (right) than in the original SD blend (left)

Discussion

The simplicity of this workflow is one of its greatest strengths. By applying a single deconvolution operator based on well-derived wavelets, significant improvements can be made to the spectral decomposition output without reprocessing the seismic data. This makes the method ideal for interpreters working with legacy datasets or time-limited projects.

However, it's important to note that the technique does not "recover" information lost due to attenuation; rather, it enhances the interpretability of what remains. The method relies on an accurate wavelet and assumes a relatively consistent wavelet across the seismic volume, which may not hold in all geological settings.

In highly heterogeneous areas or zones with significant lateral wavelet variation, a more localized or adaptive deconvolution approach may yield better results. Nevertheless, even using a consistent wavelet across shallow and deep reservoirs—as used in this study—provides substantial benefits in most practical cases.

Conclusions

This study presents a practical and efficient approach to enhancing the utility of spectral decomposition in seismic interpretation workflows through deterministic post-stack deconvolution. The method addresses

a persistent challenge in seismic analysis: limited frequency bandwidth in deeper or attenuated sections of seismic data. By increasing frequency resolution, even by modest margins, interpreters can derive greater geological insights from existing data without the need for extensive reprocessing or advanced computational resources. This advancement directly contributes to more accurate subsurface characterization, which is especially critical in complex or marginal reservoirs.

Key Outcomes and Interpretational Benefits

Post-stack deconvolution significantly enhanced the usable frequency range of the seismic data, extending the upper limit from approximately 60 Hz to 75 Hz. This improvement enabled clearer and more detailed spectral decomposition, especially in zones where traditional methods are typically limited by energy loss and poor resolution. In deeper stratigraphic intervals, these enhancements translated into more continuous reflectors and improved delineation of geological boundaries, offering a better understanding of depositional environments.

The ability to resolve thinner beds and subtle facies variations without the need for pre-stack data makes this method particularly attractive. It improves interpreter confidence, enables more robust geological models, and supports high-resolution reservoir characterization. Importantly, the workflow is non-disruptive, requiring no significant changes to interpretation software or infrastructure. As a plug-and-play solution, it integrates seamlessly into existing projects and datasets, encouraging widespread adoption across exploration and development teams.

Broader Implications for Exploration and Development

The implications of improved frequency content extend beyond basic image clarity. Enhanced vertical resolution allows geoscientists to better distinguish between depositional units, identify stratigraphic pinch-outs, and more precisely map lateral facies transitions. These capabilities are especially relevant in settings where small-scale geological features exert a significant influence on reservoir performance, such as fluvial, deltaic, or turbidite systems.

In exploration contexts, this technique helps reduce uncertainty during early-stage prospect evaluation. By making more subtle features visible, it aids in more accurate volumetric estimation, improving resource classification and economic forecasting. In development settings, particularly in mature fields, it helps optimize infill drilling strategies and improve recovery by identifying previously overlooked compartments or bypassed pay zones.

The benefits are not restricted to conventional plays. In unconventional settings, such as tight sands or shale reservoirs, better resolution of stratigraphic heterogeneities can improve geomechanical modeling, support fracture network prediction, and inform hydraulic stimulation designs. Thus, the workflow supports a broad spectrum of geological environments and operational needs.

Future Work and Development Opportunities

Several promising avenues exist for extending the utility of this method. One direction involves the development of spatially adaptive deconvolution operators that can respond to lateral changes in wavelet characteristics and seismic noise. Such adaptive methods could improve performance in geologically complex areas, including faulted or structurally deformed regions, where conventional operators may struggle to provide consistent results.

Another area of development is the integration of this workflow with pre-stack seismic attributes. Combining post-stack deconvolution with amplitude variation with offset (AVO) analysis, elastic impedance inversion, or anisotropy detection could further enhance its interpretive value. These integrations could help geoscientists discriminate between lithology and fluid effects, refine direct hydrocarbon indicators, and improve overall reservoir risk assessments.

Finally, automating parts of the deconvolution workflow using machine learning may streamline parameter selection and reduce interpreter bias. Leveraging training datasets to optimize deconvolution parameters could allow the approach to scale more efficiently across multiple assets or basins. Continued research in these directions will ensure that post-stack deconvolution remains a vital tool in the evolving landscape of data-driven geoscience.

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